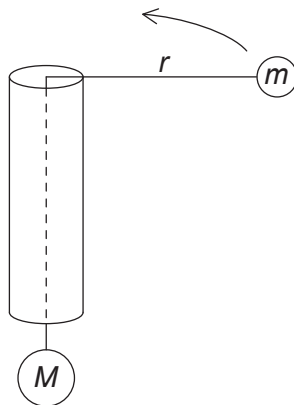


PAPER 1 / QUESTION 7

8820-6501 / original question 20

7. Mass M is attached to one end of a string. The string is passed through a hollow tube and mass m is attached to the other end. Friction between the tube and string is negligible.



Mass m travels at constant speed v in a horizontal circle of radius r . What is mass M ?

- A. $\frac{mv^2}{r}$
- B. mv^2rg
- C. $\frac{mgv^2}{r}$
- D. $\frac{mv^2}{gr}$